Differential Equations: C0743
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Lecture 3: Higher order DE (II)

4.1 Variation of Parameters

4.1.1 Linear second-order DEs

Consider a second-order linear DE in the standard form

$$y'' + P(x)y' + Q(x)y = f(x), \tag{1}$$

where $P(x)$, $Q(x)$, and $f(x)$ are continuous on some common interval I.

If

$$y_c = c_1 y_1 + c_2 y_2$$

is the complementary function for (1), variation of parameters consists of finding a particular solution of (1) of the form

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x). \tag{2}$$

where the parameters c_1 and c_2 have been replaced with functions u_1 and u_2, or *variable parameters*.

To find u_1 and u_2, first, substitute (2) in (1) to obtain (see remark below)

$$\begin{aligned} y_1u_1' + y_2u_2' &= 0 \\ y_1'u_1' + y_2'u_2' &= f(x) \end{aligned} \qquad \text{or} \qquad \begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ f(x) \end{bmatrix}. \tag{3}$$

Next, solve for u_1' and u_2' using Cramer's rule:

$$u_1' = \frac{W_1}{W}, \quad u_2' = \frac{W_2}{W} \tag{4}$$

$$\text{where} \quad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}, \quad W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}.$$

Finally, find u_1 and u_2 by integrating u_1' and u_2' in (4).

The general solution of the equation is then

$$y = y_c + y_p.$$

Remark 1. The equations in (3) are obtained as follows. Substituting (2) and its first and second derivatives into (1) and grouping terms gives

$$\begin{aligned} y'' + P(x)y' + Q(x)y &= u_1\overbrace{[y_1'' + P(x)y_1' + Q(x)y_1]}^{\text{zero}} + u_2\overbrace{[y_2'' + P(x)y_2' + Q(x)y_2]}^{\text{zero}} \\ &\quad + \frac{d}{dx}[y_1u_1' + y_2u_2'] + P[y_1u_1' + y_2u_2'] + y_1'u_1' + y_2'u_2' \\ &= \frac{d}{dx}[y_1u_1' + y_2u_2'] + P[y_1u_1' + y_2u_2'] + y_1'u_1' + y_2'u_2' = f(x). \end{aligned} \tag{5}$$

Since we seek to determine two unknown functions u_1 and u_2, we need two equations. We can obtain these two equations from (5) by making the further assumption that the functions u_1 and u_2 satisfy $y_1u_1' + y_2u_2' = 0$, which reduces (5) to $y_1'u_1' + y_2'u_2' = f(x)$. These two equations in blue form the system of equations (3).

Example 1. Solve $y'' + y = \frac{1}{4}\csc x$.

Solution: Since the roots of the auxiliary equation $m^2 + 1 = 0$ are $m_1 = i$ and $m_2 = -i$, the complementary function is $y_c = c_1 \cos x + c_2 \sin x$. Using $y_1 = \cos x$, $y_2 = \sin x$, and $f(x) = \frac{1}{4}\csc x$, we obtain

$$W(\cos x, \sin x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1,$$

$$W_1 = \begin{vmatrix} 0 & \sin x \\ \frac{1}{4}\csc x & \cos x \end{vmatrix} = -\frac{1}{4}, \quad W_2 = \begin{vmatrix} \cos x & 0 \\ -\sin x & \frac{1}{4}\csc x \end{vmatrix} = \frac{1}{4}\frac{\cos x}{\sin x}.$$

Integrating $\quad u_1' = \frac{W_1}{W} = -\frac{1}{4} \quad$ and $\quad u_2' = \frac{W_2}{W} = \frac{1}{4}\frac{\cos x}{\sin x} \quad$ yields

$$u_1 = -\frac{1}{4}x \qquad \text{and} \qquad u_2 = \frac{1}{4}\ln|\sin x|.$$

Thus from (2) a particular solution is

$$y_p = -\frac{1}{4}x\cos x + \frac{1}{4}(\sin x)\ln|\sin x|.$$

The general solution of the equation is

$$y = y_c + y_p = c_1\cos x + c_2 \sin x - \frac{1}{4}x\cos x + \frac{1}{4}(\sin x)\ln|\sin x|.$$

This equation represents the general solution of the differential equation on, say, the interval $(0, \pi/2)$.

4.1.2 Higher-order equations

The above method can be generalized to linear nth-order equations that have been put into the standard form

$$y^{(n)} + P_{n-1}(x)y^{(n-1)} + \cdots + P_1(x)y' + P_0(x)y = f(x). \tag{6}$$

If

$$y_c = c_1y_1 + c_2y_2 + \cdots + c_ny_n$$

is the complementary function for (6), then a particular solution is

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x) + \cdots + u_n(x)y_n, \tag{7}$$

where $u_1, \ldots, u_n$ are determined by integrating the system of n linear first-order differential equations

$$\begin{array}{ccccccccc} y_1u_1' & + & y_2u_2' & + & \cdots & + & y_nu_n' & = & 0 \\ y_1'u_1' & + & y_2'u_2' & + & \cdots & + & y_n'u_n' & = & 0 \\ \vdots & & & & & & & & \vdots \\ y_1^{(n-1)}u_1' & + & y_2^{(n-1)}u_2' & + & \cdots & + & y_n^{(n-1)}u_n' & = & f(x). \end{array} \tag{8}$$

or

$$\begin{bmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \\ \vdots \\ u_n' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ f(x) \end{bmatrix} \tag{9}$$

The first $n-1$ equations in this system are assumptions introduced to simplify the equation obtained by substituting $y_p = u_1(x)y_1(x) + \cdots + u_n(x)y_n(x)$ into (6), ultimately reducing it to the nth equation of the system.

To integrate (9) [or, equivalently (8)], first use Cramer's rule to get

$$u_k' = \frac{W_k}{W}, \quad \text{for } k = 1, 2, \ldots, n, \tag{10}$$

where W is the Wronskian of $y_1, \ldots, y_n$ and W_k is the determinant obtained by replacing the kth column of the Wronskian with the column consisting of the right-hand side of (9) [or, equivalently (8)]. Next, for $k = 1, \ldots, n$, find u_k by integrating (10).

The general solution of (6) is then

$$y = y_c + y_p.$$

Example 2. Solve the following third-order DE by variation of parameters: $y''' + y' = \tan x$.

Solution:

$$y = c_1 + c_2\cos x + c_3\sin x - \ln|\cos x| - \sin x \ln|\sec x + \tan x|.$$

4.2 Undetermined coefficients

This method can be used to solve the following type of problem:

Solve

$$L[y] \equiv ay'' + by' + cy = f(x), \tag{11}$$

where $a \neq 0$, b and c are constants, and $f(x)$ is a special type of function.

4.2.1 Simple case

Let us assume $f(x)$ has the simple form

$$a_1\, p(x)\, e^{a_2 x}\, \cos(a_3 x) \qquad \text{or} \qquad a_1\, p(x)\, e^{a_2 x}\, \sin(a_3 x),$$

where $a_1,\ a_2,\ a_3$ are constants and $p(x)$ is a polynomial.

- Compute the **complementary part** of the solution to (11): $y_c = c_1 y_1 + c_2 y_2$,
- Compute $f(x),\ f'(x),\ f''(x),\ \ldots$. Write down all the different terms which arise (through the product rule), ignoring constant factors and plus/minus signs:

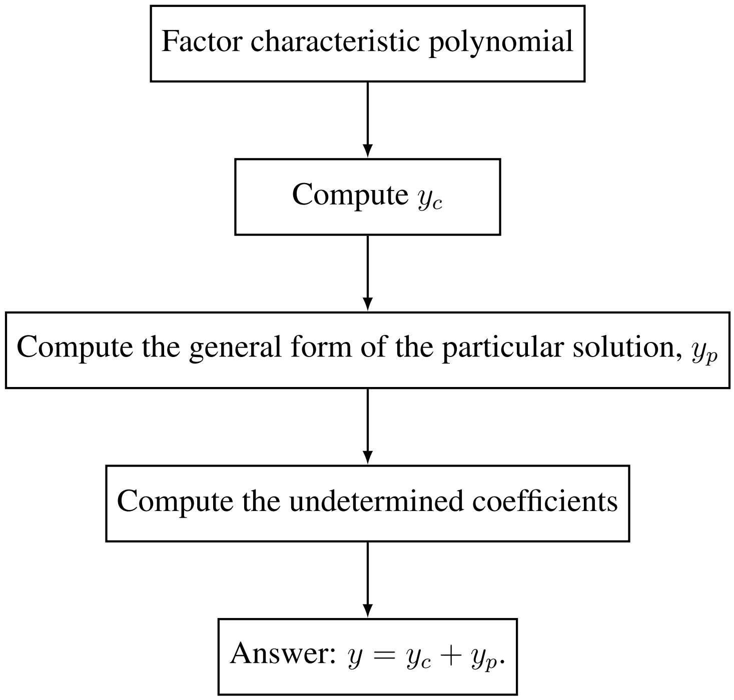
  $$t_1(x),\ t_2(x),\ \ldots,\ t_k(x). \tag{12}$$

  If any of these agrees with y_1 or y_2 then multiply them all by x. (If after this, any of them still agrees with y_1 or y_2 then multiply them all again by x.)
- Let y_p be a linear combination of these functions (your "guess"):

  $$y_p = A_1 t_1(x) + \cdots + A_k t_k(x). \tag{13}$$

  This is called the **general form of the particular solution** (when you have not solved for the constants A_i). The A_i's are called **undetermined coefficients**.
- Substitute y_p into (11) and solve for $A_1, \ldots, A_k$.
- Let $y = y_c + y_p = c_1 y_1 + c_2 y_2 + y_p$. This is the **general solution** to (11). If there are any initial conditions for (11), then solve for $c_1,\ c_2$ now.

Diagramatically:



Example 3. Solve $y'' + y = xe^x$.

Solution:

- The characteristic polynomial is $m^2 + 1 = 0$, which has $\pm i$ for roots. The complementary function is therefore $y_c = c_1 \cos x + c_2 \sin x$.
- We compute $f(x) = xe^x$, $f'(x) = xe^x + e^x$, $f''(x) = xe^x + 2e^x$, $\ldots$. We have $f^{(n)}(x) = xe^x + ne^x$ for $n \geq 0$. They are all linear combinations of
$$f_1 = xe^x, \; f_2(x) = e^x.$$
None of these agrees with $y_1 = \cos x$ or $y_2 = \sin x$, so we do not multiply by x.
- Let y_p be a linear combination of these functions
$$y_p = A_1 xe^x + A_2 e^x.$$
- Differentiating y_p and substituting the results into the DE gives, after regrouping,
$$(2A_1)xe^x + (2A_1 + 2A_2)e^x = xe^x.$$
Since this equation has to be an identity, the coefficients of like powers of x must be equal.
That is, $\quad 2A_1 = 1, \quad 2A_1 + 2A_2 = 0.$
Solving this system of equations leads to the values $A_1 = \frac{1}{2}, \quad A_2 = -\frac{1}{2}$.
- The general solution is: $y = y_c + y_p = c_1 \cos x + c_2 \sin x + \frac{1}{2} xe^x - \frac{1}{2} e^x$.

4.2.2 Non-simple case

Suppose now that

$$f(x) = f_1(x) + f_2(x) + \cdots + f_k(x),$$

where each $f_j(x),\ j = 1,\ 2,\ \ldots,\ k$ is one of the "simple" functions of the previous subsection. We have two options:

1. Split up the problem by solving each of the k problems

   $$L[y] = ay'' + by' + cy = f_j(x).$$

   Suppose

   $$L[y] = f_j(x) \quad \text{has solution} \quad y = y_j(x).$$

   The solution to (11) is then $y = y_1 + y_2 + \cdots + y_k$ (by the superposition principle).

2. Proceed as in the simple case but with the following slight revision:
   - Compute the **complementary part** of the solution to (11): $y_c = c_1y_1 + c_2y_2$,
   - Compute $f(x),\ f'(x),\ f''(x),\ \ldots$. List all the different terms that arise (through the product rule), ignoring constant factors and plus/minus signs:

     $$t_1(x),\ t_2(x),\ \ldots,\ t_k(x). \tag{14}$$

   - Group these terms into their families. Each family is determined from its parent(s)– which are the terms in $f(x) = f_1(x) + f_2(x) + \cdots + f_k(x)$ they arose from by differentiation. For instance, if $f(x) = x\cos(2x) + e^{-x}\sin x$, then the terms obtained from differentiation, ignoring constants and plus/minus signs, are

     $$\begin{array}{ll} x\cos(2x),\ x\sin(2x),\ \cos(2x),\ \sin(2x) & \qquad \text{(from } x\cos(2x)\text{)} \\ e^{-x}\sin x,\ e^{-x}\cos x & \qquad \text{(from } e^{-x}\sin x\text{).} \end{array}$$

     Thus, there are two families in this example:

     $$\{x\cos(2x),\ x\sin(2x),\ \cos(2x),\ \sin(2x)\}$$

     constitutes a *family*, where $x\cos(2x)$ is the *parent* and $x\sin(2x)$, $\cos(2x)$ and $\sin(2x)$ are its *children*; and

     $$\{e^{-x}\sin x,\ e^{-x}\cos x\}$$

     constitutes a second *family*, where $e^{-x}\sin x$ is the *parent* and $e^{-x}\cos x$ is its *child*.

     If any term in a family agrees with y_1 or y_2 then multiply the *entire family* by x. (If, after this, any of the terms still agrees with y_1 or y_2 then multiply them all again by x.)

- Let y_p be a linear combination of these functions (your "guess"):

$$y_p = A_1\tilde{t}_1(x) + \cdots + A_\ell\tilde{t}_\ell(x). \tag{15}$$

  This is called the **general form of the particular solution** (when you have not solved for the constants A_i). The A_i's are called **undetermined coefficients**.

- Substitute y_p into (11) and solve for $A_1, \ldots, A_\ell$.
- Let $y = y_c + y_p = c_1y_1 + c_2y_2 + y_p$. This is the **general solution** to (11). If there are any initial conditions for (11), then solve for $c_1,\ c_2$ now.

4.3 Green's functions

We now analize the **response** or **output** $y(x)$ of

$$y'' + P(x)y' + Q(x)y = f(x) \tag{16}$$

to either initial conditions or boundary conditions directly in terms of the **forcing function** or **input** $f(x)$. In this way the response of the system can quickly be analyzed for different forcing functions. We assume throughout this section that the coefficient functions $P(x), Q(x)$, and $f(x)$ are continuous on some common interval I.

4.3.1 Initial-value problems

Here we show that the solution $y(x)$ of the second order IVP

$$y'' + P(x)y' + Q(x)y = f(x), \quad y(x_0) = y_0, \quad y'(x_0) = y_1, \tag{17}$$

can be expressed as the superposition of two solutions:

$$y(x) = y_h(x) + y_p(x), \tag{18}$$

where $y_h(x)$ is the solution of the associated homogeneous DE with nonhomogeneous initial conditions

$$y'' + P(x)y' + Q(x)y = 0, \quad y(x_0) = y_0, \quad y'(x_0) = y_1, \tag{19}$$

(here at least one of y_0 or y_1 is assumed to be nonzero, for if both are zero, the solution of the IVP is $y = 0$), and $y_p(x)$ is the solution of the nonhomogeneous DE with homogeneous initial conditions

$$y'' + P(x)y' + Q(x)y = f(x), \quad y(x_0) = 0, \quad y'(x_0) = 0. \tag{20}$$

Green's function

If $y_1(x)$ and $y_2(x)$ form a fundamental set of solutions on the interval I of the associated homogeneous form of (16), then a particular solution of the nonhomogeneous equation (16) on the interval I can be found by variation of parameters. Namely,

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x), \tag{21}$$

$$u_1'(x) = \frac{\begin{vmatrix} 0 & y_2(x) \\ f(x) & y_2'(x) \end{vmatrix}}{W(x)} = -\frac{y_2(x)f(x)}{W(x)}, \qquad u_2'(x) = \frac{\begin{vmatrix} y_1(x) & 0 \\ y_1'(x) & f(x) \end{vmatrix}}{W(x)} = \frac{y_1(x)f(x)}{W(x)} \tag{22}$$

If $x,\ x_0 \in I$ then

$$\begin{aligned} y_p(x) &= y_1(x)\int_{x_0}^{x} \frac{-y_2(t)f(t)}{W(t)}dt + y_2(x)\int_{x_0}^{x} \frac{y_1(t)f(t)}{W(t)}dt \\ &= \int_{x_0}^{x} \frac{-y_2(t)y_1(x)}{W(t)}f(t)dt + \int_{x_0}^{x} \frac{y_1(t)y_2(x)}{W(t)}f(t)dt \end{aligned} \tag{23}$$

where

$$W(t) = W(y_1, y_2)(t) = \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix}$$

Therefore

$$y_p(x) = \int_{x_0}^{x} G(x,t)f(t)dt \tag{24}$$

$$G(x,t) = \frac{y_1(t)y_2(x) - y_1(x)y_2(t)}{W(t)} \tag{25}$$

$$= \frac{\begin{vmatrix} y_1(t) & y_2(t) \\ y_1(x) & y_2(x) \end{vmatrix}}{W(t)} \tag{26}$$

The function $G(x,t)$ is called the **Green's function** for the DE (16).

Remark 2. A Green's function (25) depends only on the fundamental solutions $y_1(x)$ and $y_2(x)$ of the associated homogeneous DE for (16) and not on the forcing function $f(x)$. Hence all linear second-order DEs (16) with the same left-hand side but with different forcing functions have the same Green's function.

Theorem 1 (Solution of the IVP (20))**.** *The function $y_p(x)$ defined in* (24) *is the solution of the initial-value problem* (20).

Theorem 2 (Solution of the IVP (17))**.** *If $y_h(x)$ is the solution of the IVP* (19) *and $y_p(x)$ is the solution* (24) *of the IVP* (20) *on the interval I, then*

$$y(x) = y_h(x) + y_p(x) \tag{27}$$

is the solution of the IVP (17).

It follows from (27) that the response $y(x)$ of a physical system described by the IVP (17) can be separated into two different responses:

$$\begin{aligned} y(x) = y_h(x) &\quad \text{response of system due to initial conditions } y(x_0) = y_0,\ y'(x_0) = y_1 \\ + & \\ y_p(x) &\quad \text{response of system due to the forcing function } f \end{aligned} \tag{28}$$

Example 4. Solve the IVP

$$y'' + 4y = x, \quad y(0) = 0, \quad y'(0) = 0.$$

Solution: Let's begin by building the Green's function for the DE. The two linearly independent solutions of $y'' + 4y = 0$ are $y_1(x) = \cos 2x$ and $y_2(x) = \sin 2x$. Moreover $W(y_1, y_2)(t) = 2$. From (25) we thus get

$$G(x,t) = \frac{\cos 2t \sin 2x - \cos 2x \sin 2t}{2} = \frac{1}{2}\sin 2(x-t).$$

Since $x_0 = 0$ and $f(t) = t$, from (24) we find that a solution of the IVP is

$$\begin{aligned} y_p(x) &= \frac{1}{2}\int_0^x t \sin 2(x-t)dt \\ &= \frac{1}{4}x - \frac{1}{8}\sin 2x. \end{aligned}$$

Example 5. Solve the IVP

$$y'' + 4y = \sin 2x, \quad y(0) = 1, \quad y'(0) = -2.$$

Solution: First, we solve

$$y'' + 4y = 0, \quad y(0) = 1, \quad y'(0) = -2.$$

Applying the initial conditions to the general solution $y(x) = c_1 \cos 2x + c_2 \sin 2x$ of the homogeneous DE, we get $c_1 = 1$ and $c_2 = -1$. Thus,

$$y_h(x) = \cos 2x - \sin 2x.$$

Next, we solve

$$y'' + 4y = \sin 2x, \quad y(0) = 0, \quad y'(0) = 0.$$

The Green's function is

$$G(x,t) = \frac{1}{2}\sin 2(x-t)$$

(why?). With $f(t) = \sin 2t$, we find from (24) that the solution to this second problem is

$$y_p(x) = \frac{1}{2}\int_0^x \sin 2(x-t)\sin 2t dt.$$

Finally, in view of (27), the solution of the original IVP is

$$y(x) = y_h(x) + y_p(x) = \cos 2x - \sin 2x + \frac{1}{2}\int_0^x \sin 2(x-t)\sin 2t dt.$$

4.3.2 Boundary-value problems

A BVP for a second-order DE involves conditions on $y(x)$ and $y'(x)$ that are specified at two different points $x = a$ and $x = b$. We shall use the following homogeneous boundary conditions:

$$A_1 y(a) + B_1 y'(a) = 0 \tag{29}$$

$$A_2 y(b) + B_2 y'(b) = 0, \tag{30}$$

where $A_i,\ B_i,\ i = 1,\ 2$ are constants.

We want to find an integral solution $y_p(x)$ that is analogous to (24) for nonhomogeneous BVPs of the form

$$\begin{aligned} y'' + P(x)y' + Q(x)y &= f(x) \\ A_1 y(a) + B_1 y'(a) &= 0 \\ A_2 y(b) + B_2 y'(b) &= 0. \end{aligned} \tag{31}$$

We assume that $P(x), Q(x)$, and $f(x)$ are continuous on $[a, b]$, and that the homogeneous problem

$$\begin{aligned} y'' + P(x)y' + Q(x)y &= 0 \\ A_1 y(a) + B_1 y'(a) &= 0 \\ A_2 y(b) + B_2 y'(b) &= 0 \end{aligned}$$

has only the trivial solution $y = 0$. This latter assumption is sufficient to guarantee that a unique solution of (31) exists and is given by an integral $y_p(x) = \int_a^b G(x,t)f(t)dt$, where $G(x,t)$ is a Green's function.

Suppose $y_1(x)$ and $y_2(x)$ form a fundamental set of solutions on the interval $[a,b]$ of the associated homogeneous form of (31) and that $x \in [a,b]$. Instead of integrating the derivatives in (22) over the same interval, as in the construction of (23), here we integrate the first equation in (22) on $[b,x]$ and the second equation in (22) on $[a,x]$:

$$u_1(x) = -\int_b^x \frac{y_2(t)f(t)}{W(t)}dt, \quad \text{and} \quad u_2(x) = \int_a^x \frac{y_1(t)f(t)}{W(t)}dt \tag{32}$$

From (32), a particular solution $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$ of the DE is

$$\begin{aligned} y_p &= y_1(x)\int_x^b \frac{y_2(t)f(t)}{W(t)}dt + y_2(x)\int_a^x \frac{y_1(t)f(t)}{W(t)}dt \\ &= \int_a^x \frac{y_2(x)y_1(t)}{W(t)}f(t)dt + \int_x^b \frac{y_1(x)y_2(t)}{W(t)}f(t)dt \end{aligned} \tag{33}$$

Therefore

$$y_p(x) = \int_a^b G(x,t)f(t)dt \tag{34}$$

$$G(x,t) = \begin{cases} \dfrac{y_1(t)y_2(x)}{W(t)}, & a \le t \le x \\ \dfrac{y_1(x)y_2(t)}{W(t)}, & x \le t \le b. \end{cases} \tag{35}$$

The function $G(x,t)$ is called the **Green's function** for the boundary value problem (31). $G(x,t)$ is a continuous function of x on the interval $[a,b]$.

We have the following theorem.

Theorem 3 (Solution of the BVP (31))**.** *Let $y_1(x)$ and $y_2(x)$ be linearly independent solutions of*

$$y'' + P(x)y' + Q(x)y = 0$$

on $[a,b]$, and suppose $y_1(x)$ satisfy (29)*, $A_1y_1(a) + B_1y_1'(a) = 0$, and $y_2(x)$ satisfy* (30)*, $A_2y_2(b) + B_2y_2'(b) = 0$. Then the function $y_p(x)$ defined in* (34) *is a solution of the BVP* (31).

Example 6. Solve the following BVP where k is a positive even number and F is a nonzero real constant:

$$y'' + k^2 y = F, \quad y'(0) = 0, \quad y(\pi) = 0.$$

Solution: The solutions of the associated homogeneous equation

$$y'' + k^2 y = 0$$

are

$$y_1(x) = \cos kx \quad \text{and} \quad y_2(x) = \sin kx,$$

where $y_1(x)$ satisfies $y_1'(0) = 0$ ($a = 0, A_1 = 0, B_1 = 1$ in (31)) and $y_2(x)$ satisfies $y_2(\pi) = 0$ ($b = \pi, A_2 = 1, B_2 = 0$ in (31)). The Wronskian is

$$W(y_1, y_2)(x) = k,$$

and so from (35) we see that the Green's function for the BVP is

$$G(x,t) = \begin{cases} \dfrac{1}{k} \cos kt \sin kx, & 0 \le t \le x \\ \dfrac{1}{k} \cos kx \sin kt, & x \le t \le \pi. \end{cases}$$

It follows from Theorem 3 that the solution to the BVP is given by (34), with $a = 0$, $b = \pi$, and $f(t) = F$:

$$\begin{aligned} y_p(x) &= F \int_0^\pi G(x,t) dt \\ &= F \cdot \frac{1}{k} \sin kx \int_0^x \cos kt dt + F \cdot \frac{1}{k} \cos kx \int_x^\pi \sin kt dt \\ &= \frac{F}{k^2}(1 - \cos kx). \end{aligned}$$

4.4 Systems of Linear DEs

Example 7. Solve by elimination

$$\begin{aligned} \frac{dx}{dt} &= 9y \\ \frac{dy}{dt} &= 4x \end{aligned}$$

Solution: Apply $\dfrac{d}{dt}$ to the first (or second) equation of the sytem and continue from there.